

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (ISE) (Credit System) Degree Examinations**

January 2023

20IS504 - DATA MINING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I**Marks BT* CO* PO***

1. a) Consider the following 2D dataset.

	A1	A2
X1	1.5	1.7
X2	2	9
X3	1.6	1.8
X4	1.2	1.5
X5	1.5	1.0

Consider the data as 2D data points. Given a new data point $x = (1.4, 1.6)$ as a query, rank the database points based on similarity with the query using Euclidean distance, Manhattan distance, supremum distance and cosine similarity.

10 L*3 1 2

- b) Consider the below dataset which shows the list of events each candidate is participating.

Participation ID	Events
P1	Singing, Rangoli, Drama
P2	Singing, Rangoli
P3	Singing, Dance, Painting
P4	Painting, Dance
P5	Painting, Drama
P6	Singing, Dance, Painting

i) Find all frequent itemset using Apriori algorithm.

ii) Find the strong association rules.

Assume minimum support = 33%, minimum confidence = 60%.

10 L3 2 2

2. a) With examples, explain types of attributes w.r.t to data mining.

10 L2 1 1

- b) Using the data for age given: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(i) Plot an equal-width histogram of width 10.

(ii) Sketch examples of each of the following sampling techniques: SRSWOR, SRSWR, cluster sampling, and stratified sampling. Use samples of size 5 and the strata "youth," "middle-aged," and "senior."

10 L3 1 2

3. a) Explain various ways to handle missing values in a dataset.

10 L2 1 1

- b) Consider the below dataset

Transaction ID	Items bought
T100	f, a, c, d, g, i, m, p
T200	a, b, c, f, l, m, o
T300	b, f, h, j, o
T400	b, c, k, s, p
T500	a, f, c, e, l, p, m, n

Find the frequent itemset using FP growth algorithm.

10 L3 2 2

Unit – II

4. a) Explain bayes theorem with the relevant formula.
b) Explain briefly how does classification work.
c) Explain decision tree algorithm with pseudocode.

05 L2 3 1

05 L2 3 1

10 L2 3 1

P.T.O.

5. a) Apply the ID3 algorithm to the training data in the table.

Training	fever	vomiting	diarrhea	shivering	Classification
d_1	no	no	no	no	healthy (H)
d_2	average	no	no	no	influenza (I)
d_3	high	no	no	yes	influenza (I)
d_4	high	yes	yes	no	salmonella poisoning (S)
d_5	average	no	yes	no	salmonella poisoning (S)
d_6	no	yes	yes	no	bowel inflammation (B)
d_7	average	yes	yes	no	bowel inflammation (B)

Identify the root node.

10 L3 3 2
10 L2 4 2

- b) Explain different metrics for evaluating classifier performance.

6. a) Let's consider a simple dataset comprised of 10 data samples:

Type of family structure	Age group	Income status	Will they buy a car?
Nuclear	Young	Low	Yes
Extended	Old	Low	No
Childless	Middle-aged	Low	No
Childless	Young	Medium	Yes
Single Parent	Middle-aged	Medium	Yes
Childless	Young	Low	No
Nuclear	Old	High	Yes
Nuclear	Middle-aged	Medium	Yes
Extended	Middle-aged	High	Yes
Single Parent	Old	Low	No

Given three inputs—for example, Single Parent, Young, and Low—compute the probability of these people buying a car, using naïve bayes algorithm.

10 L3 3 2
10 L2 4 2

- b) Explain rule induction using a sequential covering algorithm.

Unit – III

7. a) Explain the requirements of cluster analysis.
b) Cluster the following eight points (with (x, y) representing locations) into three clusters:

10 L2 5 1

A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is defined as-

$$P(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

Use K-Means Algorithm to find the three cluster centers after the second iteration.

10 L3 5 2

8. a) Explain various clustering methods with its advantage and disadvantages

10 L2 5 1

- b) Consider a single vector $D = \{18, 22, 43, 42, 27, 25\}$. Draw a dendrogram using complete link clustering method.

10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
